Jace Innis

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RESEARCH EXPERIENCE

Research interests: phytoplankton ecology, the biological carbon pump, marine snow production and export, and harmful algal blooms.

Marine Snow Particle Classification — *Ocean Flux Program (OFP) Image Time Series*

Apr 2025 – Present

- Trained and deployed a YOLOv8 segmentation model on 20+ years of marine snow imagery; segmented and classified 5+ million particles across 75,000+ images into biological categories.
- Built automated Python/R workflows for image preprocessing, prediction, and post-processing, utilizing the ASU Sol cluster (Slurm batch scheduling, CUDA-accelerated model training).

HAB (Red Tide) Early-Warning Model — *CSU Monterey Bay — Palacios Remote Sensing Lab / UROC*

Jul 2020 – May 2022

- Designed a CNN-based early-warning model to predict harmful algal blooms using spatial/temporal predictors from MODIS satellite ocean-color data.
- Presented HAB early-warning model results at CSU Monterey Bay research symposium.

NSF REU Research Intern — *Bigelow Laboratory for Ocean Sciences — Mentor: Dr. Nick Record*

Aug 2021 – May 2022

- Developed TensorFlow/Keras deep learning models to categorize thousands of bacterial genomes and marine metagenomic sequences.
- Poster & Oral Presentation: presented genomic classification results at Bigelow Laboratory.

TECHNICAL SKILLS

Oceanographic sensor testing, calibration, and installation (CTD, ADCP, gliders, dissolved-oxygen and acoustic sensors); oscilloscope/multimeter component-level diagnostics; mechanical and electrical systems troubleshooting (SMAW/TIG/MIG welding and fabrication); Python, R; Windows and Linux/UNIX command-line administration; small-boat and research-vessel operations.

PROFESSIONAL EXPERIENCE

Marine Science Technician II — *BIOS (R/V Atlantic Explorer), Bermuda* **Apr 2024 – Present**

- Deploy and recover CTD rosettes, moorings, AUVs, and gliders using hydraulic and electric winches, A-frames, and hand lines.
- Calibrate and repair CTDs, ADCPs, dissolved-oxygen sensors, and met-ocean suites to the component level using oscilloscopes, multimeters, and precision soldering.
- Administer and repair shipboard Linux/Windows data collocation and transfer systems, as well as satellite network communications.
- Author and maintained SOPs, calibration records, and cruise reports to standardize operations across the technician team.
- Train visiting scientists and junior technicians on shipboard instrumentation and systems.

Coastal Data Specialist — *Running Tide, Seattle, WA (Remote)* **Aug 2023 – May 2024**

- Automated Python/R pipelines for multi-sensor buoy data array, including QA/QC into a real time data dashboard.
- Built GIS models and dashboards for coastal restoration and habitat-suitability analysis.

Coastal Resource Management Volunteer — *U.S. Peace Corps, Ilocos Sur, Philippines*
Dec 2022 – Sep 2023

- Organised small-boat marine habitat surveys across 5+ km of coastal waters with local government units and fisherfolk communities.
- Authored technical reports and recommendations to local government stakeholders.

Hull Technician / Work Center Supervisor / Fire Marshal — *U.S. Navy — USS Howard (DDG 83)*
Mar 2013 – Jun 2017

- Supervised a 5-person engineering work center, managing an $85,000+ annual maintenance budget and equipment calibration schedule.
- Fabricated replacement hardware and piping to ~1mm tolerances via SMAW/TIG/MIG welding.
- Served as Fire Marshal, leading weekly damage-control and firefighting training for 100+ personnel.

EDUCATION

Bachelor of Science, Marine Science (Cum Laude, GPA 3.65), Minor in Statistics
California State University, Monterey Bay | May 2022

Relevant coursework: Experimental Design & Analysis, Statistical Computing in R, Practical Computing for Scientists, Time-Series Analysis, Oceanography, Quantitative Analysis.

- Teaching Assistant (Jan 2021 – May 2022): led applied statistics lab sessions in R (tidyverse, R Markdown), graded projects and analysis reports, and tutored students in hypothesis testing, ANOVA, and regression.
- Student Researcher (Jul 2020 – May 2022), Palacios Remote Sensing Lab (Dr. Sherry Palacios) and UROC: conducted satellite remote sensing research on harmful algal bloom prediction.

CERTIFICATIONS & ADDITIONAL QUALIFICATIONS

- AAUS Scientific Diver / Research SCUBA Diver (Active)
- STCW Basic Safety Training (Personal Survival, Firefighting, First Aid, Personal Safety & Social Responsibilities)
- ASU Laboratory & Supercomputer (HPC) Safety Training
- U.S. Navy Damage Control & Firefighting Specialist

REFERENCES

Dr. Nick Record
Senior Research Scientist
Bigelow Laboratory for Ocean Sciences
NSF REU Research Mentor
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Rod Johnson
Principal Investigator, BATS
Bermuda Institute of Ocean Sciences
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Dr. Sherry Palacios
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